

Title: Guidelines for Determining Brain Death/Death by Neurologic Criteria

Policy #: PR-12

Purpose:

- A. To provide procedures for practitioners to assist in making the determination that a patient is deceased by neurologic criteria.
- B. To ensure consistency among practitioners regarding the practices and procedures related to the determination of death by neurologic criteria.

Policy Statement:

A. In the past, cessation of circulatory function was the basis for diagnosing death, whether the initial failure occurred in the brain, the heart, lungs or elsewhere in the body. Because circulation and respiration can be maintained for a significant period by means of mechanical ventilation and other interventions, despite loss of brain function, we also recognize as legally dead those individuals whose loss of brain function is complete and permanent. Thus there are two different circumstances in which we recognize that death has occurred:

1. Permanent cessation of respiration and circulation.
2. Permanent cessation of all brain function, including the brainstem, resulting in coma, brainstem areflexia, and apnea in the setting of an adequate stimulus, with artificial maintenance of cardiopulmonary function (neurologic assessment of brain function required), known as brain death or death by neurologic criteria (BD/DNC).

B. In using these procedures, the following points are important:

1. These procedures assume the use of clinical judgment is necessary in applying and interpreting tests of brain function. Thus, these procedures do not replace the physician's judgment in individual cases, as BD/DNC is a clinical diagnosis.
2. Evaluation for the diagnosis of BD/DNC should be considered for all patients who are suspected of having permanent cessation of all brain function. BD/DNC testing should be initiated in all appropriate patients regardless of eligibility or family intent for organ donation. Although frequent communication is imperative to prepare family or surrogate decision makers for the suspected diagnosis of BD/DNC, explicit consent from them is not required. However, the patient's appropriate surrogate may refuse BD/DNC testing in circumstances where a final decision has been made to pursue comfort care and the withdrawal of life support.
3. All deaths must be reported to the New England Donor Services. See Beth Israel Deaconess Medical Center Organ Donation Policy (Policy # PR-09) for instructions.
4. In the absence of consideration of organ donation, patients when pronounced dead should have artificial support systems discontinued. While permission of the family to disconnect support systems after BD/DNC declaration is not required (and efforts to obtain such permission is not recommended), it is appropriate and respectful that family be kept informed of such proceedings.
5. The fact that BD/DNC is death should be explained clearly to the family, with the assistance of Interpreter Services, when applicable. Emphasis should be on the fact that the patient is dead despite appearances. Therefore, terms such as "disconnecting life support" or "letting him/her die" are inaccurate and inappropriate.

and should not be used.

Guideline(s) for Implementation:

- A. Determination of BD/DNC using the criteria identified herein should be the responsibility of the attending physician in consultation with critical care, neurology, or neurosurgery attending if the attending physician is not a critical care, neurology or neurosurgery attending. All patients whose severe neurologic injury may result in BD/DNC should be referred to the New England Donor Services (NEDS) for evaluation of organ donation potential. This should happen prior to BD/DNC testing and prior to conversations regarding options for comfort care and withdrawal of mechanical supportive devices to ensure that NEDS has adequate time for a thorough evaluation and screening of the patient's chart, medical history, and donor registration status. BIDMC staff will work collaboratively with NEDS staff to evaluate these patients, and when potential donors are identified, ensure families are made aware of their options to donate in the most sensitive manner possible.

Note: Physicians with patients on organ donation wait lists and physicians who may participate in organ transplant or organ recovery should not be involved in the determination of BD/DNC. The treating physicians should request consultation with a neurologist, neurosurgeon or a critical care physician.

1. An examination must be carried out by, or under the direct supervision of, a Neurology, Neurosurgery, or Critical Care attending. While one examiner is required per BIDMC policy, a second examination may decrease the risk of a false-positive determination (i.e. incorrectly determining BD/DNC when it is not the case) due to diagnostic error, and can be considered.

If available within the time frame dictated by clinical need, a PGY-2 or above Neurology or Neurosurgery resident, or Neurocritical Care fellow should conduct the examination under direct supervision of the responsible attending. Advanced practice providers (APP) who practice critical care may also conduct examinations under supervision of critical care attending staff.

The attending physician or fellow/resident/APP must complete the Brain Death Declaration Checklist (Attachment A) and place in the paper chart. This checklist provides and documents in the patient's medical record the means by which the diagnosis of BD/DNC was established. The attending's signature is required. The checklist becomes part of the medical record.

2. The declaration of BD/DNC and the explanation to the family is the responsibility of the attending physician who declares BD/DNC. Discussions of organ donation will occur following a collaborative plan with the New England Donor Services, typically by a trained requestor from NEDS. It is not appropriate for the medical team to discuss organ donation with the family before a collaborative plan has been developed with New England Donor Services. In instances where family initiates a discussion of organ donation, acknowledge there may be a possibility, but defer further discussion until a collaborative plan has been established with NEDS.

- B. Appropriate diagnostic and therapeutic procedures should be performed to document the irreversibility of the patient's condition prior to the diagnosis of BD/DNC. BD/DNC may only be declared in the appropriate medical setting:

There must be a known cause for BD/DNC. This may be clinical or by neuroimaging evidence of an acute CNS catastrophe that is compatible with clinical BD/DNC.

And

The patient should not have a condition that may simulate total lack of nervous system function, but be potentially reversible. This would include but not be limited to the following conditions:

1. Hypothermia below 36 degrees Celsius (91 degrees Fahrenheit)
2. Total paralysis due to neuromuscular blockage or peripheral nerve disease
3. Severe metabolic disease, including, but not limited to:

Hepatic, renal, hyperosmolar, hypercarbic, hypoxic, hypotensive or septic encephalopathy. In instances where an acute CNS catastrophe is evident on neuroimaging and clinical examination supports BD/DNC, ancillary testing may be considered to confirm BD/DNC and rule out severe metabolic disease.

Notes:

- a. Hypotension: clinicians should maintain SBP $\geq$ 100 mm Hg and MAP $\geq$ 75 mm Hg prior to evaluation for BD/DNC
- b. Clinicians should ensure that severe metabolic derangements affecting the central nervous systems are excluded and adequately corrected prior to evaluation for BD/DNC. The list of tests and associated reference ranges below are practical suggestions based on consensus, and thus there are no definitive cut-off laboratory values that exclude evaluation for BD/DNC. Therefore, clinical context and judgment are required.

- i. Ammonia $\geq$ 75 μ mol/L
- ii. BUN $\geq$ 75 mg/dL
- iii. Calcium $<$ 7 mg/dL or $>$ 11 mg/dL
- iv. Glucose $<$ 70mg/dL or $>$ 300 mg/dL
- v. Magnesium $<$ 1.5mg/dL or $>$ 4mg/dL
- vi. Potassium $<$ 3 mmol/L or $>$ 6 mmol/L
- vii. Sodium $<$ 130 mmol/L or $>$ 160 mmol/L
- viii. pH $<$ 7.3 or $>$ 7.5
- ix. (If clinical concern for endocrinopathy) Total T4 $<$ 3mg/dL or $>$ 30mg/dL, free T4 $\leq$ 0.3 ng/dL or $>$ 5 ng/dL

4. Post-ictal state following severe or prolonged seizures for up to 24 hours
5. Sedative or depressive drugs

Notes:

- Ensure phenobarbital level (if administered or if known medication) $<$ 5ug/mL or below the lower limit for detection

- Ensure blood alcohol level is < 80 mg/dL

6. Patients with primary infratentorial injury: in cases where patients have an isolated brainstem pathology, ancillary testing (see acceptable methods below) is recommended to demonstrate whole brain death even in the clinical examination and apnea test are consistent with BD/DNC.

C. Timing of the examination after therapeutic hypothermia or targeted temperature management following cardiac arrest should be delayed for a minimum of 24-hours after return of core body temperature to 36 degrees Celsius, or longer depending on timing of administration of medication(s) that could depress the central nervous system. A longer duration than 5 half-lives of elimination may be needed as therapeutic hypothermia prolongs medication clearance, and concomitant hepatic or renal dysfunction is common. If uncertainty exists regarding sufficient time for medication clearance, a complete clinical examination and apnea test should be performed for BD/DNC, and an ancillary study should be obtained. No ancillary test has 100% sensitivity or specificity, and ancillary testing should not be pursued before obtaining the most complete and reliable clinical examination possible.

D. In addition to the criteria outlined above, the following neurologic criteria should be established:

1. Cerebral Unresponsiveness as demonstrated by deep coma with no evidence of withdrawal away from painful stimuli (pressure at the supraorbital area, temporomandibular joint, sternum, OR finger OR toe nail beds). *Spinal level movement [stereotyped movement toward a painful stimulus, occasionally bicycling leg movements or even sitting up spontaneously (the Lazarus Syndrome) may be seen] does not preclude the diagnosis of BD/DNC, but decerebrate (stereotyped extension of all four limbs) or decorticate (stereotyped flexion of upper extremities and extension of lower extremities) posturing in response to pain does.*
2. Pupillary Reflex Unresponsiveness. Previously healthy pupils should be unreactive to bright light bilaterally to support the diagnosis of BD/DNC.
3. Other Cranial Nerves should be assessed for reflexes. There should be no corneal reflex, oculoccephalic reflex (Doll's eyes), facial movement or bulbar function such as gagging or coughing with tracheal stimulation (deep suctioning)
4. Oculovestibular testing should be performed (with patient's head elevated to 30 degrees and clear external auditory canals). Slow irrigation of 10 ml/min for 5 min of ice cold water into each ear (trauma patients should be assessed for integrity of the tympanic membrane via bedside otoscopic examination) should produce no eye movement. Even if the oculoccephalic reflex is unable to be assessed due to concern for cervical spine/skull base integrity, oculovestibular testing can be used to determine BD/DNC without the need for ancillary testing. This is acceptable only if all other elements for BD/DNC are satisfied (e.g. patients with ocular trauma may have unreliable pupillary light reflexes, and ancillary testing should be pursued).
5. Apnea testing should be performed in the following manner:
 - a. Prior to the test, the ventilator shall be adjusted to give a PaCO₂ between 35 and 45 mm Hg. The patient should be off all sedative or paralytic agents, and hypotension should be corrected as best as possible.
 - b. Patient shall be placed on FiO₂=100% for 5 minutes. Baseline vital signs and blood gas analysis shall be recorded.

- c. The patient's thorax and abdomen should be exposed (to visualize for respiratory effort), the patient removed from the ventilator and an oxygen catheter supplying 6-8 liters of oxygen per minute inserted into the trachea.
- d. The patient must be observed for a period of ten minutes for the presence of:
 1. Spontaneous respirations (which indicates residual brainstem function)
 2. Visible cyanosis or $\text{SaO}_2 < 85$ (which should cause the test to be aborted).
- e. At the time of observation of any of the above events or completion of ten minutes of observation an arterial blood gas should be drawn and the patient placed back on the mechanical ventilator at the previous settings.

If the PaCO_2 is ≥ 60 mm Hg and ≥ 20 mm Hg above pre-apnea test baseline level, and arterial pH < 7.3 , and no spontaneous respirations or significant changes in blood pressure or heart rate have occurred after observation, the requirements for the apnea test have been fulfilled and support the diagnosis of BD/DNC.

For patients with chronic hypercarbia and known baseline PaCO_2 , the apnea testing target is $\text{PaCO}_2 \geq 60$ mm Hg and ≥ 20 mm Hg above premorbid baseline level, and arterial pH < 7.3 .

E. Ancillary testing:

If there is any question clinically about the origin of the unresponsiveness or, if any portion of the clinical evaluation cannot be completed, appropriate ancillary testing must be employed. An ancillary test should NOT be used as a substitute for the clinical determination of BD/DNC, except when a clinical determination cannot be safely or accurately performed. The neurologic examination and apnea test must be performed to the fullest extent possible, and findings must be consistent with BD/DNC before ancillary testing is performed.

Typical indications include, but are not limited to:

- Inability to correct metabolic derangements
- Underlying medical conditions, such as fracture of cervical spine/skull base/orbit, severe facial injuries
- Inability to determine whether motor movements are spinally mediated
- Inability to perform apnea testing due to hemodynamic instability
- Knowledge/suspicion of chronic hypercarbia without knowledge of baseline PaCO_2

There are two accepted ancillary tests at BIDMC:

- a. Nuclear medicine cerebral blood flow study (SPECT), OR
- b. Catheter-based conventional cerebral angiography

Notes on ancillary testing:

- a. Establishing total lack of cerebral blood flow with radiological or radionuclide imaging is a gold standard of declaring BD/DNC and may be considered definitive when the patient has any of the

conditions indicated in Part B or, when the clinical examination cannot be completed due to technical limitations. The time of death when an ancillary test is used is the *date and time of the signature of the **attending physician** interpreting the ancillary test in the medical record.* [MR angiography can falsely indicate no blood flow when blood flow is reduced and is not an adequate substitute, CT Angiography has not been validated for use as an ancillary test.]

b. EEG only determines the state of cortical activity and is useful if it shows activity, e.g. in a patient who is deafferented. However, lack of EEG activity is not sufficient grounds for determining BD/DNC as it does not test hypothalamic or brainstem function.

c. Transcranial Doppler Ultrasound (TCD) can be limited by the absence of insonation windows in certain individuals.

If cessation of all brain function is confirmed by examination and appropriate testing, the patient may be declared deceased by neurologic criteria.

F. Special populations (pediatric patients, patients on extracorporeal membrane oxygenation, pregnant patients)

1. Pediatric patients

Patients <18 years of age should be evaluated for BD/DNC by **2** examiners, with a **minimum of 12 hours** separating the examinations. This includes apnea testing, which should be performed at each evaluation.

For blood pressure/MAP targets, SBP and MAP should be maintained $\geq$ fifth percentile for age.

For neonatal patients (< 6 months of age), consultation with Boston Children's Hospital Neurology should be obtained as per the BIDMC Neonatal ICU protocol.

2. Patients on extracorporeal membrane oxygenation (ECMO)

When performing apnea testing for patients on ECMO, preoxygenation should be provided through both the membrane lung and the ventilator.

To achieve adequate rise in PaCO₂, exogenous CO₂ can be added to the ECMO circuit and/or the sweep gas flow rate can be titrated down to 0.2 – 1L/min.

In patients on venoarterial ECMO, arterial blood should be sampled from both the distal arterial line and the ECMO circuit post-oxygenator. Values from both locations must meet the pH and PaCO₂ targets.

For ancillary testing, there is limited guidance in the literature and in major society recommendations. Key elements for consideration include the following:

1. 4-vessel angiography: Contrast opacification of vessels may be difficult to interpret in patients on VA-ECMO support, since these

patients have 2 competing circulations (systemic and ECMO) that mix within the aorta.

2. Radionuclide testing: The ECMO circuit increases the volume of distribution of the radionuclide tracer, but the distribution is affected less by flow rates. Thus, the nuclear medicine study is the preferred modality for ancillary testing in VA-ECMO patients.
3. Transcranial dopplers: Interpretation should be done with caution as TCDs rely on pulsatile flow, and may be inaccurate in ECMO patients.

3. Pregnant patients

Pregnancy itself is not a contraindication to BD/DNC evaluation. There should be multidisciplinary discussions with experts in maternal-fetal medicine, neonatology, neurology before, during, and after declaration of BD/DNC.

Workflow for declaration of death by neurologic criteria

- A. *Prior* to testing for and pronouncement of BD/DNC, consultation and referral to the New England Donor Services (800-446-6362) must be initiated.
- B. If the neurological findings indicate BD/DNC and all other criteria for BD/DNC have been met, this information will be clearly recorded in the patient's medical record by the critical care or neurology or neurosurgery attending or neurology or neurosurgery resident with attending signature.

If, in consultation with the critical care attending, neurologist, or neurosurgeon, it is determined that confirmatory tests are required to declare BD/DNC, they should be performed.

If examination and ancillary testing, if required, support the diagnosis of BD/DNC, the attending or designated resident shall enter a note in the medical record indicating that the patient is declared dead. The criteria used for the determination of death, the date, time and signature of the attending should be included in this note. In addition, the Brain Death Declaration Checklist should be completed and entered into the patient's medical record. Ancillary test images must be interpreted and the report signed by an attending radiologist prior to declaration.

- C. *After* BD/DNC has been declared and documented as noted above, and *after* consultation with the New England Donor Services Donation Coordinator has determined no potential for organ donation exists, the ventilator and other support systems may be disconnected by the respiratory therapist, registered nurse, resident or attending physician upon the written order of the attending physician or designee.
- D. Established hospital policies and pertinent laws regarding autopsy and review of death by a medical examiner must be considered prior to disposition of the body.

Attachment A: Brain Death Declaration Checklist

Vice President Sponsor: Mary LaSalvia, MD, CMO

Approved By:

☒ **Medical Executive Committee: 5/20/26**

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Original Date Approved: 10/02 Next Review Date:

5/29

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Eliminated:

**References: Attachment A: Brain Death Checklist Neurology 2023;
101(24): 1112-1132**

Declaration of Death by Neurologic Criteria Checklist

**Must be included in Medical Record

Attending Physician _____ Date of Examination: ____ / ____ / ____
Examining Physician _____ ☐ New England Donor Services contacted (1-800-446-6362)
Assisting House Staff or Provider: _____ Second Examining Physician (if applicable) _____

I. PREREQUISITES (All "NO" or "NON-APPLICABLE" responses must be explained in Section III)

1. Clinical or neuro-imaging evidence of CNS catastrophe compatible with clinical diagnosis of brain death
☐ Yes ☐ No
2. Absence of complicating medical conditions:
 - Absence of Hypothermia (core temperature 96.8 degrees Fahrenheit / 36 degrees Celsius or greater).
☐ Yes ☐ No
 - Absence of paralysis due to neuromuscular blockage or severe peripheral nerve disease (e.g. Guillain Barre Syndrome, myasthenia gravis or drugs) ☐ Yes ☐ No
 - Absence of severe metabolic encephalopathy (such as hepatic, renal, hyperosmolar, hypercarbic, hypoxic, hypotensive or septic encephalopathy) ☐ Yes ☐ No
 - Absence of post-ictal state (severe seizures in previous 24 hours) ☐ Yes ☐ No
 - Absence of sedative or depressive drugs ☐ Yes ☐ No
 - If any of the above cannot be excluded, absence of cerebral blood flow by approved ancillary test has been confirmed ☐ Yes ☐ No

II. The patient lacks evidence of function of the brain in the following neurological bedside tests:

- a. CEREBRAL UNRESPONSIVENESS as demonstrated by deep coma with **no** evidence of withdrawal away from painful stimuli (pressure at the supraorbital area, or at the temporomandibular joint, over the sternum, or on the finger or toe nail beds). Spinal level movement (withdrawal into the painful stimulus, some stereotyped axial movements such as bicycling leg movements or even sitting up spontaneously [Lazarus syndrome]) does not preclude the diagnosis of brain death, but decerebrate or decorticate posturing does.
☐ Yes ☐ No
- b. ABSENT PUPILLARY LIGHT REFLEXES: Previously healthy pupils should be 3-9 mm and non-reactive when tested with a bright light.
[Note: pupils of death are midposition and fixed. The fixed and dilated pupils seen immediately after death are due to circulating catecholamines released agonally. Once the catecholamines are metabolized which takes 15-60 minutes in most cases, the pupils will come down to about 3 mm. Small pupils compared to baseline, however, always indicate brain function].
☐ Yes ☐ No ☐ Non-Applicable
- c. ABSENT CRANIAL NERVE MOTOR FUNCTION: There should be **no** facial movement, **no** corneal reflex and **no** response to gag or coughing during tracheal stimulation (deep suctioning).
☐ Yes ☐ No
- d. ABSENT EYE MOVEMENTS/OCULOVESTIBULAR FUNCTION: There should be **no** eye movements either spontaneously or when the head is rotated to the side or up and down, or with caloric testing. (With the patient's head elevated to 30 degrees, the auditory canal should be cleared and then infused with ice water which should be in contact with the tympanic membrane for at least 5-minutes. There should be **no** eye movement.). If oculocephalic reflex testing is unable to be performed due to concern for cervical spine/skull base injury, oculovestibular testing may be performed instead without the need for ancillary testing, as long as other elements of the neurological examination are able to be tested and are consistent with brain death. ☐ Yes ☐ No ☐ Non-Applicable

- e. **APNEA TESTING:** There should be no breathing response during apnea testing performed as follows
- ☐ Prior to the test, the ventilator adjusted to give PaCO₂ between 35 and 45 mmHg and 100% O₂ administered.
 - ☐ Vital signs and blood gas analysis is recorded after 5 minutes.
 - ☐ The patient is removed from the ventilator and an oxygen catheter supplying 6-8 liters per minute is inserted into the trachea.
 - ☐ The patient's thorax and abdomen should be exposed to visualize any respiratory effort. Patient observed for 10 minutes for spontaneous respirations. At the completion of ten minutes observation, an arterial blood gas is drawn and the patient placed back on the ventilator at the previous settings.
 - ☐ If the PaCO₂ is > 60 mmHg, or PaCO₂ is greater than 20mmHg above the patient's pre-testing baseline (or pre-morbid baseline in known CO₂ retainers), and pH < 7.3, and **no** spontaneous respirations were noted, apnea test is complete and support the diagnosis of brain death.

Note: The test should be aborted if there is visible cyanosis or SaO₂ <85% due to incomplete oxygenation

APNEA TEST SUCCESSFULLY COMPLETED ☐ Yes ☐ No*

*If apnea test is unable to be completed, absence of cerebral blood flow has been confirmed by radionuclide or conventional catheter based cerebral angiographic imaging and the radiology report has been signed by the attending radiologist. ☐ Yes

II. MANDATORY DOCUMENTATION AND PROCEDURES

- ☐ The neurological findings and other criteria for brain death have been recorded in the patient's medical record by the critical care, neurology, neurosurgery attending or the neurology resident. The time of death is dated, timed and signed by the attending physician.
- ☐ Brain Death Declaration checklist has been completed and placed in the patient's medical record.
- ☐ New England Donor Services (NEDS) contacted. If potential for organ donation exists, removal of support equipment delayed as necessary. Family members have been fully apprised of all options regarding donation by NEDS- if not applicable, please check here: ☐ explain in Section III below.
- ☐ If NEDS coordinator determines no potential for organ donation, support systems may be removed by the respiratory therapist, nurse, resident or attending physician upon the written order of the attending physician or designee.
- ☐ Policies for post-mortem examination and referral of cases to medical examiner followed.

III. NOTES/COMMENTS

Signature of physician/provider completing checklist

Date and time completed

Printed name of physician/provider

Signature of second examining physician/provider (if applicable)

Date and time completed

Printed name of physician/provider

Signature of attending physician

Date and time

Printed name of attending physician